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Teachers' Verbal Lexical Explanation for Second Language Vocabulary Learning: A Meta-Analysis

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Abstract

This article reports a meta-analysis of studies on the effects of teachers' verbal lexical explanation (TVLE), as one type of lexical focus on form, on second language vocabulary learning. The dataset for this meta-analysis included 14 studies, representing a total of 36 independent samples (N = 3,304). The results of this study reveal that TVLE resulted in more vocabulary gains than an absence of TVLE, at posttest and delayed posttest. In addition, explanation in the first language was more effective than explanation in the second language at posttest and delayed posttest. We further identified a range of potential moderators that may influence the effects of TVLE. In particular, we found that the overall effectiveness of TVLE was greater when the degree of vocabulary knowledge was measured through recall tests rather than through recognition tests in the long term.

Keywords: bilingual approach; lexical focus on form; meta-analysis; vocabulary instruction; vocabulary learning

Introduction

Diverse approaches have been suggested and examined for effective second language (L2) vocabulary instruction, with one of these approaches being operationalized as lexical focus on form (lexical FonF, henceforth). Lexical FonF refers to the method that "attends to lexical items … since these lexical items are necessary for the completion of a communicative or an authentic language task" (Laufer, 2006, p. 150). In addition to decontextualized and intentional vocabulary learning approaches (see Webb, Yanagisawa, & Uchihara, 2020, for a meta-analysis), and incidental vocabulary learning approaches (for a meta-analysis see Uchihara, Webb, & Yanagisawa, 2019), some lexical FonF techniques have been examined in

L2 vocabulary intervention research (Barclay & Schmitt, 2019; Laufer, 2009; and see Yanagisawa & Webb, 2021, for a meta-analysis). Teachers' verbal lexical explanation (TVLE, henceforth) in meaning-focused activities (e.g., listening and reading comprehension) has been considered one of the effective types of lexical FonF in both first language (L1; e.g., Biemiller & Boote, 2006; Brett, Rothlein, & Hurley, 1996; Elley, 1989) and L2 research (e.g., J. H. Lee & Levine, 2020; Tian & Macaro, 2012). Although TVLE may vary in terms of its content (see Appendix S1 in the Supporting Information online), it essentially involves providing a definition (or a translation equivalent in a L1) for a lexical item, with its purpose lying in the enhancement of learners' comprehension of target texts and acquisition of the target vocabulary.

Although L2 research has seen an increasing number of qualitative classroom studies examining TVLE (e.g., Rolin-Ianziti & Brownlie, 2002; Tian & Hennebry, 2016), as well as experimental studies (e.g., J. H. Lee & Levine, 2020; Tian & Macaro, 2012; Zhang & Graham, 2020), there has been no meta-analytic effort to analyze the findings of these studies comprehensively. Given that TVLE is considered the oral equivalent of glossing (i.e., written lexical information about the target words), but with a more fleeting nature, a recent meta-analysis of studies that have examined the effects of glossing on L2 vocabulary learning from reading (Yanagisawa, Webb, & Uchihara, 2020) has led us to conduct a meta-analysis on the effects of TVLE. Thus, our aim in the present meta-analysis was to collect and analyze previous experimental studies on the effects of TVLE on L2 vocabulary learning. Our analysis also provides methodological implications for lexical FonF research and pedagogical implications for vocabulary instruction by L2 teachers.

In this meta-analysis, the primary inquiry is whether TVLE (as one type of lexical FonF) leads to larger vocabulary gains than a lack thereof. Furthermore, we aimed to examine the impact of a range of moderators on the effectiveness of TVLE, including which language (i.e., the explanation being in the L1 or L2) is more effective for L2 vocabulary learning. As proficiency level has been suggested as a key variable that influences the effects of TVLE (e.g., J. H. Lee & Macaro, 2013; Macaro, Guo, Chen, & Tian, 2009), we also investigated how learners with different L2 proficiency levels differentially benefit from TVLE. We further aimed to identify a range of other moderators related to TVLE (e.g., the ratio of target words to the total words of the passage, testing format), with some of them having been identified and discussed as potential variables in previous empirical studies and others being selected based on previous meta-analyses on L2 glossing (Kim, Lee, & Lee, 2020; Yanagisawa et al., 2020; Yun, 2011). With these moderators, we aim to provide an in-depth examination of the extent to which these variables influence the effects of TVLE.

Background Literature

The Effects of Teachers' Verbal Lexical Focus on Form

The assumption that TVLE promotes learners' vocabulary development has been initially examined in L1 classroom research in relation to teachers' brief explanations of words during story-reading sessions (e.g., Biemiller & Boote, 2006; Brett et al., 1996; Elley, 1989). This type of explanation can be suggested to be an example of lexical FonF, in that teachers attempt to shift children's attention, briefly, to target words included in a story.

In instructed L2 learning contexts, where the teacher is considered one of the most important sources of language input (J. H. Lee, 2020), it has been proposed that learners would develop their L2 vocabulary knowledge through TVLE (Macaro et al., 2009; Meara, Lightbown, & Halter, 1997). Indeed, TVLE of L2 words has been observed in a wide range of meaning-focused tasks (e.g., Rolin-Ianziti & Brownlie, 2002; Tian & Hennebry, 2016). In these lessons, teaching vocabulary has not been the primary aim of instruction, but it has been integrated into more meaning-focused tasks (e.g., reading or listening comprehension), with TVLE serving as lexical FonF (Laufer, 2006, 2009). For example, previous studies on the effects of TVLE (e.g., J. H. Lee & Levine, 2020; Tian & Macaro, 2012) reported that the instructor implemented listening comprehension activities, with the instruction goal being the learners' comprehension of the listening passages in the target language. Then, when difficult words—that is, target words that were inserted into these passages by the researchers and that were important in comprehending the meaning of these passages—emerged in the passage, the instructor briefly shifted the learners' attention from the listening passages to these words, explaining them verbally.

Among the several theoretical perspectives on L2 learning, two frameworks may directly support the value of TVLE. The first is the "noticing" hypothesis (Robinson, Mackey, Gass, & Schmidt, 2012; Schmidt, 2001), which proposes that some degree of attention is required for learning to occur (see Kormos, 2019, for a review of different perspectives on how noticing influences L2 vocabulary learning). Within our context, teachers' brief focus on target words may raise the learners' level of attention, which in turn may increase the likelihood that they will learn these words. The second framework is expressed as the "involvement load" hypothesis (Laufer & Hulstijn, 2001), which suggests that the involvement load imposed on learners by the target task may influence the degree of subsequent vocabulary learning (see Yanagisawa & Webb, 2021, for a meta-analytic synthesis). Again, with this context in mind, the challenging words that need to be understood in order to comprehend a target passage may

present some cognitive load for the learners, which would not only motivate them to comprehend the meaning of these words but also promote the learning of these words through TVLE.

Evidenced Moderators of the Effectiveness of TVLE

In this section, we introduce a range of variables that have been identified as key influences on the effectiveness of TVLE, along with some relevant references to L2 vocabulary research more generally.

Language Choice of Teachers' Verbal Lexical Explanation

Our initial perusal of existing literature revealed that most studies on TVLE have compared the effects of teachers' use of different languages (i.e., the L1 or L2). As another example of lexical FonF, a recent meta-analysis of glossing (i.e., the written equivalent of TVLE; Kim et al., 2020) has shown that the effect of language of glossing has been addressed intensively in research; it revealed that L1 glossing generally results in more vocabulary gains than its L2 counterpart, suggesting a strong argument for the use of L1 input in L2 vocabulary learning.

This argument is supported by Jiang's (2000, 2004) three-stage model of L2 vocabulary acquisition. According to this model, the first stage of L2 vocabulary acquisition is concerned with the registration of a new lexical entry, which only "contains form specifications and a pointer that links the word to its L1 translation" (Jiang, 2004, p. 418). At this stage, the comprehension and usage of this new L2 word (and access to its concept) are predominantly mediated via the lemma information (i.e., semantic and syntactic information) of its L1 translation, which is also transferred to the entry of the L2 word. With increasing exposure to contextualized input related to this word (i.e., in the second stage), a target lexical entry contains more L2-related lemma information, with this entry building a direct link to the concept. In the final stage, a stronger link between the L2 word and the concept is established, and reliance on L1 translation is reduced. Jiang (2004) notes that "many words may stop short of this third stage and L1 lemma mediation may become a steady state of lexical processing in advanced L2 learners" (p. 417).

Kroll and Stewart's (1994) revised hierarchical model offers a similar prediction regarding the role of the L1 in the early stage of L2 acquisition, proposing that L2 users are likely to gain access to the L2 word concept through the lexical link from the L2 word to its L1 equivalents (rather than the direct link between the L2 word and concept) at earlier stages (see Kemp & McDonald, 2021, for a recent study broadly supporting this framework). However, it should

be noted that, according to this model, the lexical links from the L2 word to its L1 equivalent do not disappear even at later stages when a more direct link between the L2 word and its concept starts to be built.

Finally, according to the dynamic systems theory (Larsen-Freeman & Cameron, 2008), which supposes an important role of initial conditions (e.g., learners' L1 system) for L2 development (Verspoor, De Bot, & Lowie, 2011), it can be assumed that "learners' existing L1 mental lexicon interacts with newly acquired L2 lexical items" (Kormos, 2019, p. 216). Such an assumption, along with the propositions of the abovementioned psycholinguistic models, suggests that learners' L1 should not be avoided in L2 vocabulary teaching and learning, at least in the initial stages. Rather, it may be more realistic to consider how to effectively engage it.

Learners' Second Language Proficiency Level

Learners' L2 proficiency level has also been considered to moderate the effects of TVLE, particularly between L1 and L2 explanations. In studies based on sociocultural theory (Lantolf, 2006; Vygotsky, 1978), the L1 is known to help lower level learners engage in interactions with others (e.g., peers, teachers), serving as an important medium for collaboration (e.g., DiCamilla & Antón, 2012). The aforementioned psycholinguistic models (Jiang, 2000, 2004; Kroll & Stewart, 1994) have proposed that learners with lower L2 proficiency levels would strongly rely on L1 translation or equivalents of a L2 word to gain access to its concept, but that they would do so less with an increase in L2 proficiency level. In view of these studies, it can be hypothesized that a L1 explanation would benefit lower level learners more than their peers with higher level L2 proficiency.

A frequently discussed variable in L2 research, along with L2 proficiency level, is learners' age. However, we decided not to include "age" as a potential moderator, because in the studies on the effects of TVLE, it was found that the participants' L2 proficiency level generally coincided with their age, with younger learners having lower levels of L2 proficiency. For example, the participants in the studies by Lugo-Neris, Jackson, and Goldstein (2010) and Song and Lee (2019) were preschoolers and were beginner-level L2 learners. Conversely, older participants in other studies (e.g., Jin & Webb, 2020; Tian & Macaro, 2012; Zhao & Macaro, 2016) belonged to pre-intermediate levels and above.

Potential Moderators

In the present meta-analysis we considered a range of moderators predicted to be relevant to the effects of TVLE in meaning-focused activities. This section will provide the rationale for including them, with these moderators being grouped into instructional features (i.e., [a] type

of TVLE, [b] input mode, [c] timing of TVLE, [d] availability of written forms of target words, [e] ratio of target words to total words, and [f] length of instruction), measurement and context (i.e., [g] testing format and [h] learning context), and [i] retention interval.

Type of Teachers' Verbal Lexical Explanation

The type of TVLE has been considered a potential moderator. Regarding teachers' explanation of L2 vocabulary, Macaro et al. (2009) noted that L2 teachers can choose from a wide range of options in L2 lexical instruction, and even fine-tune their lexical explanation to the students' level, for which Macaro et al. proposed the term "teacher as dictionary designer." In previous studies on this issue, as mentioned above, TVLE has generally referred to the definition (or equivalent in L1) of the target vocabulary. However, in some of these studies, TVLE included additional lexical information such as sample sentences that contained the target words and some semantic features of the target words (for details, see 'Type of TVLE' in Appendix S1 in the Supporting Information online).

Input Mode

Some of the previous studies on TVLE used the reading mode (e.g., J. H. Lee & Macaro, 2013; Zhao & Macaro, 2016), whereas others used the listening mode (e.g., Hennebry, Rogers, Macaro, & Murphy, 2017; Zhang & Graham, 2020). This general body of research has not often focused on direct comparisons of vocabulary acquisition in listening versus reading. However, some literature has suggested that the input mode may play a role in learners' L2 vocabulary learning. For example, some studies on incidental L2 vocabulary acquisition (e.g., Brown, Waring, & Donkaewbua, 2008; Hatami, 2017; Vidal, 2011) have directly compared the acquisition of vocabulary knowledge through reading and listening and have shown that L2 learners benefited more from the reading mode.

Timing of Teachers' Verbal Lexical Explanation

Regarding the timing of TVLE, existing literature on this topic has revealed that TVLE of a target word could be given when the word appears, as the instructor and learners read or listen to the target passages (i.e., at the comprehension phase of the lesson). Alternatively, TVLE of target words could be given after the teacher and learners finish reading or listening to the target passage (i.e., after the comprehension phase). It has been hypothesized that learners would engage in more deliberate vocabulary learning in the latter case (Hulstijn, 2013), because learners can exclusively focus on the target words.

Availability of Written Forms of Target Words

Regarding the availability of written forms of target words, teachers have the option of presenting the form of the target word on a board, on PowerPoint slides, or on word cards (e.g., Akhter, 2018; Hennebry et al., 2017), or without such resources (e.g., Jin & Webb, 2020). We hypothesized that the former would lead to higher gains, because learners are given additional input.

Ratio of Target Words to Total Words

In L2 vocabulary research (e.g., Hu & Nation, 2000; Schmitt, Jiang, & Grabe, 2011; van Zeeland & Schmitt, 2013), it has been proposed that some level of coverage of the target text is required to achieve an adequate comprehension thereof, implying that low lexical coverage of the target text may hamper comprehension. Based on such a proposition, we hypothesized that the ratio of target words to the total number of words in target passages may influence L2 vocabulary learning through TVLE, in that the greater the ratio (more target words relative to length of passage) would result in less learning than a lower ratio.

Length of Instruction

The length of the instruction process is another moderator in the instructional features included in the present meta-analysis. It is possible to predict that when the target treatment has already been a part of the participants' curriculum or their instructor's pedagogical approach prior to the research, the number of sessions in the experimental intervention may not then significantly moderate the effect of this treatment. We hypothesized that the number of sessions might not be a determining variable in terms of L2 vocabulary learning in this meta-analysis, as the participants of the sampled studies are likely to have been given TVLE during reading or listening comprehension activities in their own L2 learning contexts.

Testing Format

With vocabulary knowledge being considered multidimensional (Nation, 2013), previous meta-analyses of L2 vocabulary learning (e.g., H. Lee, Warschauer, & Lee, 2019; Yanagisawa et al., 2020) have included the different types of vocabulary knowledge as one of the moderator variables. Although a wide range of testing formats are available, studies on this topic have predominantly employed two testing formats: passive recall (i.e., a test taker is given a target word and asked to recall its meaning) and passive recognition (i.e., a test taker is given a target word and asked to choose the correct meaning among the given options). According to Laufer and Goldstein's (2004) hierarchy of the strength of the form–meaning link related to the knowledge of meaning, passive recognition knowledge is likely to be acquired earlier than passive recall knowledge, which "entails certain productive skills that go

beyond meaning comprehension" (Laufer & Aviad-Levitzky, 2017, p. 733; see González-Fernández & Schmitt, 2020, for a similar proposition related to the order of acquisition of different lexical knowledge components). The addition of this moderator enables us to examine the extent to which TVLE supports L2 learners' development of recall and recognition knowledge differentially.

Learning Context

Another potential moderator is learning context. Language learning contexts can be categorized in terms of the status of the target language within the context, that is, as L2 or foreign language (FL) learning, with L2 referring to the language spoken in the wider community outside the classroom (e.g., English in the United States) and FL referring to a language mostly learned through formal classroom language instruction (e.g., Spanish in China). Although teachers' language use has traditionally been examined in L2 or FL contexts, it has recently been examined in English-medium instruction contexts (Macaro, 2018) and in content and language integrated learning contexts (Ruiz de Zarobe & Jiménez Catalán, 2009). The latter two types of context differ from L2 or FL contexts, in that lessons are designed to teach subjects (e.g., mathematics or science) through the target language. In this study, we focused only on L2 and FL contexts. In addition, it was revealed that more than 90% of the studies included in the meta-analysis were conducted in FL contexts; therefore, this moderator was not further considered in our statistical analyses (see Appendix S1 in the Supporting Information online for some qualitative description of this moderator across the sampled studies).

Retention Interval

Research on L2 vocabulary instruction has measured vocabulary learning outcomes at both posttest and delayed posttest, with the latter being implemented to examine to what extent the target intervention is effective in the longer term. TVLE research has also followed this trend, with intervals between intervention and delayed posttesting ranging from one week (Jin & Webb, 2020) to four and a half weeks (Tian & Macaro, 2012; and for a meta-analysis examining the effects of different retention intervals in L2 practice more generally, see Kim & Webb, 2022). With such variation across the included studies, retention interval was considered as a moderator to estimate the long-term effects of TVLE in a more systematic way.

The Present Study

Based on the previous literature, we aimed to address the following research questions:

1. To what extent does TVLE bring about larger L2 vocabulary gains than a lack of TVLE in the context of meaning-focused activities?
2. To what extent do the identified moderators (i.e., L2 proficiency, TVLE type, input mode, timing of TVLE, availability of written forms of target words, ratio of target words to total words, length of instruction, testing format, and retention interval) vary across the included studies, and how are they related to the effects of TVLE on L2 vocabulary learning?
3. To what extent does TVLE in the L1 bring about larger L2 vocabulary gains than TVLE in the L2 in the context of meaning-focused activities?
4. To what extent are the identified moderators (i.e., L2 proficiency, TVLE type, input mode, timing of TVLE, availability of written forms of target words, ratio of target words to total words, length of instruction, testing format, and retention interval) related to the relative effects of TVLE in the L1 or L2 on L2 vocabulary learning?

Method

This section elaborates on the steps taken for our meta-analysis, including the literature search and establishment of inclusion criteria for the meta-analysis, coding, dataset construction, effect size calculation, and data analysis plan. We were able to identify 14 studies, representing a total of 36 independent samples with 3,304 participants. Among these samples, 21 of them had posttest results, and 34 had delayed posttest results.

Literature Search and Inclusion Criteria

To identify relevant empirical studies in the literature for our meta-analysis, we conducted various literature searches, including database, forward and backward, and manual searches, in accordance with practical guidelines commonly used in previous meta-analysis research in the field (e.g., Kim et al., 2020; H. Lee et al., 2019; H. Lee, Chung, Zhang, Abedi, & Warschauer, 2020). For the database searches, we searched ProQuest (including subdatasets, such as ProQuest Dissertation & Thesis Global, ERIC, and Research Library) and Web of Science (Social Sciences Citation Index, Arts & Humanities Citation Index, and Conference Proceedings Citation Index), and used the keywords search option using the following keywords for titles and abstracts: [vocabulary] ‘AND’ [teacher, instructor, ‘OR’ instruction] ‘AND’ [L1, L2, codeswitching, code-switching, ‘OR’ translation]. Further, we searched relevant journals, such as *Language Learning*, *Language Teaching Research*, *The Modern Language Journal*, *Language Learning Journal*, *Studies in Second Language Acquisition*, and *ELT Journal*, conducting the same keyword searches to identify any further articles not found in our earlier database searches. Once identified through the abovementioned sources, each

article was utilized for a forward search (i.e., exploring articles citing the identified articles after they had been published) and a backward search (i.e., checking references in the identified articles).

As shown in Figure 1, a total of 153 studies were identified through the database, journal, and forward and backward searches. The full text of each study was reviewed to assess eligibility for inclusion. Out of 153 studies, we initially excluded 140 for the following reasons. First, some were duplicate studies (k = 7; e.g., several unpublished dissertations were published later, and we included the latter in the present study). Second, the majority (k = 130) were excluded for not meeting the inclusion criteria (see below). Third, three studies did not include the necessary descriptive statistics for effect size calculations. We contacted the authors in order to acquire data if available and later received a response, with the necessary information, from the authors of one of these studies (Lugo-Neris et al., 2010). Consequently, a total of 14 studies were selected for the current meta-analysis. The numbers of independent samples for both posttest and delayed posttest will be explained in the following section. Below, we describe each inclusion criterion in detail.

Regarding the first criterion, we checked whether studies targeted the learning of vocabulary, excluding those that examined other linguistic aspects (e.g., de la Fuente & Goldenberg, 2020; Viakinnou-Brinson, Herron, Cole, & Haight, 2012). Regarding the second criterion, we only included quantitative studies where a quasi-experimental research design (i.e., including a control or a comparison group with no TVLE, though assignment to group did not have to be randomized at either the individual or class level) was used to investigate the effects of TVLE on learners' L2 vocabulary learning.

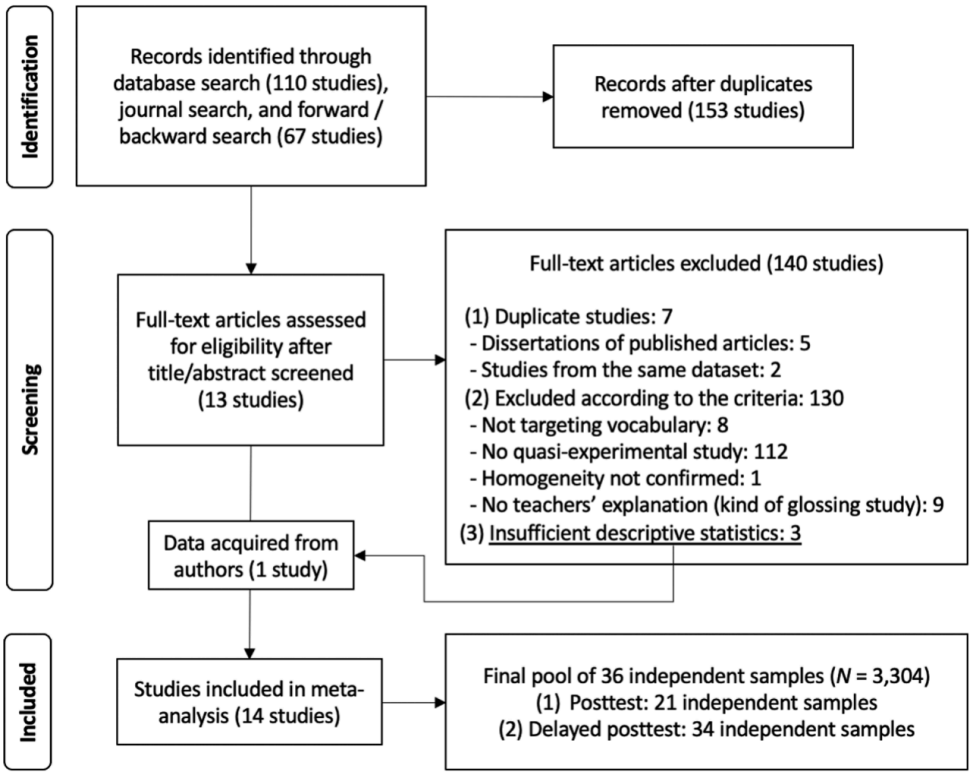


Figure 1 PRISMA flowchart (adapted from Page et al., 2021).

Regarding the third criterion, we only included studies that confirmed the homogeneity of the groups assigned to different conditions. The most popular way of doing this has been to employ a pretest of target lexical items, with some studies (J. H. Lee & Macaro, 2013; Zhang & Graham, 2020) including nontarget lexical items to measure participants' vocabulary breadth. Hennebry et al. (2017), Tian and Macaro (2012), and Zhao and Macaro (2016) administered a battery of baseline tests and ensured that the groups were comparable prior to the intervention. Random allocation of the participants on individual levels to each condition was implemented in only two studies (Hosseini & Rassaei, 2016; Tian & Macaro, 2012). In some studies (Hennebry et al., 2017; J. H. Lee & Levine, 2020; Song & Lee, 2019; Zhang & Graham, 2020), each sampled class was randomly allocated to the different conditions.

Regarding the fourth criterion, we only included those studies in which the teacher was the main source of lexical explanation (rather than glossing or learners). Finally, only studies written in English were considered.

Table 1 Coding scheme for moderator variables

Categories (4)	Moderator variables	Coding scheme
Evidenced moderators (2)	Language choice of TVLE	L1 vs. L2
	Learners' L2 proficiency level	Basic vs. Beyond basic

Categories (4)	Moderator variables	Coding scheme
Instructional features (6)	Type of TVLE	Providing definitions of vocabulary vs. Providing additional lexical information
	Input mode	Reading vs. Listening
	Timing of TVLE	After vs. During
	Availability of written forms	With forms vs. Without forms
	Ratio of target words to total words	Target words / total words (continuous)
	Length of instruction	Number of sessions (continuous)
Measurement (1)	Testing format	Recall vs. Recognition
Retention interval (1)		Number of weeks between intervention and delayed posttesting (continuous)

Note. TVLE = teachers' verbal lexical explanation; L1 = first language; L2 = second language.

Coding Scheme

The 14 identified studies were then coded to answer the research questions. We focused on the following 10 features across four categories: (a) two evidenced moderators (language choice of TVLE and learners' L2 proficiency levels); (b) six instructional features (type of TVLE, input mode, timing of TVLE, availability of written forms of target words, ratio of target words to total words, and length of instruction); (c) measurement (testing format); and (d) retention interval. As explained earlier, we did not examine the learning context as a moderator variable. See Table 1 for information related to coding for each moderator. Further details can be found in the Supporting Information online: Appendix S1 provides qualitative descriptions of the identified moderators, elucidating the contexts and differences across the selected studies, Appendix S2 gives more detailed information for each of the included studies, and Appendix S4 provides a list of the full references for the included studies.

Language Choice of Teachers' Verbal Lexical Explanation

Teacher's language choice for the explanations was coded as either L1 or L2.

Learners' Second Language Proficiency Level

A wide range of instruments was adopted in the included studies and there was a lack of detailed explanation of participants' L2 proficiency levels (see Park, Solon, Dehghan-Chaleshtori, & Ghanbar, 2021, for a synthesis showing similar reporting issues across L2 research more widely). Therefore, we divided the participants in the studies into "basic" and "beyond basic," although we acknowledge that this may be an oversimplified perspective on L2 proficiency levels. Beginner-level students or children with less than 2 years of learning the L2 were coded as "basic" and the rest as "beyond basic."

Type of Teachers' Verbal Lexical Explanation

In all sampled studies, TVLE was provided either in the form of definitions (in the L1 or L2) or as L1 equivalents. In some studies, TVLE further included additional lexical information (e.g., example sentences including the target word); thus, this variable was coded as definitions only versus additional information (i.e., including definitions or L1 equivalents).

Input Mode

The studies were conducted in either reading or listening modes.

Timing of Teachers' Verbal Lexical Explanation

The time when TVLE was delivered for instruction was coded as during or after reading or listening to the target passage.

Availability of Written Forms of Target Words

We coded for whether written forms of target words were provided or not.

Ratio of Target Words to Total Words

We calculated the ratio of the target words to the total words of the listening or reading texts (when multiple texts were involved in a study, we aggregated (added) across all the texts used in the intervention). This moderator was coded as a continuous variable.

Length of Instruction

Although sessions were of different lengths, we looked at the total number of sessions of each sample and coded it as a continuous variable (e.g., one 30-minute or one 40-minute session would be coded as '1'; two 20-minute sessions would be coded as '2').

Testing Format

Each study employed either recall or recognition tests. Other types of testing formats, such as writing, were excluded. It should be noted that, although a small number of studies (e.g., J. H. Lee & Macaro, 2013; Zhao & Macaro, 2016) employed the Vocabulary Knowledge Scale (Paribakht & Wesche, 1997) as the recall test, we did not distinguish between these studies and others that did not use that test, as they largely focused on the same aspect of vocabulary knowledge (i.e., learners' ability to recall the meaning of the target words in their L1 or L2).

Retention Interval

For the delayed posttests, the number of weeks between intervention and the delayed posttest was coded as a continuous variable.

Dataset Construction

As shown in Table 2, we found that some studies had more than one independent sample. First, Hennebry et al. (2017) and J. H. Lee and Levine (2020) had six independent samples each as a result of conducting three comparisons (L1 explanation vs. L2 explanation, L1 explanation vs. control, and L2 explanation vs. control) with two different samples. Second, Akhter (2018), J. H. Lee and Macaro (2013), Lugo-Neris et al. (2010), and Song and Lee (2019) had two independent samples each. Third, Rouzbahani and Alipour (2019), Tian and Macaro (2012), Zhang and Graham (2020), and Zhao and Macaro (2016) had three independent samples each. Furthermore, we found that there was variation among the independent samples as to whether effect sizes were provided for posttest only, delayed posttest only, or both, making a total of 21 and 34 independent samples for posttest and delayed posttests, respectively, from the final pool included in this study of 36 independent samples with a total of 3,304 participants (*N*).

Furthermore, we found that our collected data include different comparisons (by input conditions) that are not intuitively interpretable when placed together. That is, about half of our calculated effect sizes relate to the relative effects of teachers' lexical explanation (regardless of different language choices) over control conditions, and the other half describe those of teachers' L1 input over its L2 counterpart. Thus, we separated these two sets of comparisons, both of which included posttest and delayed posttest results.

Table 2 Overview of the constructed dataset for the meta-analysis

Studies	Independent samples	Sample sizes	Comparisons[a]	Posttest effect sizes[b]	Delayed posttest effect sizes[b]
Akhter (2018)	[1]	54	L1 vs. L2	–	√
	[2]	52	L1 vs. L2	–	√
Camó & Pladevall Ballester (2015)	[1]	34	TVLE (L1) vs. Ctrl	√	√
Hennebry et al. (2017)	[1]	133	L1 vs. L2	–	√
	[2]	129	L1 vs. L2	–	√
	[3]	137	TVLE (L1) vs. Ctrl	–	√
	[4]	152	TVLE (L1) vs. Ctrl	–	√
	[5]	154	TVLE (L2) vs. Ctrl	–	√
	[6]	113	TVLE (L2) vs. Ctrl	–	√
Hosseini & Rassaei (2016)	[1]	24	TVLE (L1) vs. Ctrl	√	√
Jin & Webb (2020)	[1]	172	TVLE (L1) vs. Ctrl	√	√

Studies	Independent samples	Sample sizes	Comparisons[a]	Posttest effect sizes[b]	Delayed posttest effect sizes[b]
J. H. Lee & Levine (2020)	[1]	61	L1 vs. L2	√	√
	[2]	67	L1 vs. L2	√	√
	[3]	53	TVLE (L1) vs. Ctrl	–	√
	[4]	60	TVLE (L1) vs. Ctrl	–	√
	[5]	58	TVLE (L2) vs. Ctrl	–	√
	[6]	57	TVLE (L2) vs. Ctrl	–	√
J. H. Lee & Macaro (2013)[c]	[1]	443	L1 vs. L2	√	√
	[2]	286	L1 vs. L2	√	√
Lugo-Neris et al. (2010)[d]	[1]	18	L1 vs. L2	√	–
	[2]	26	L1 vs. L2	√	–
Rouzbahani & Alipour (2019)	[1]	60	L1 vs. L2	√	√
	[2]	60	TVLE (L1) vs. Ctrl	√	√
	[3]	60	TVLE (L2) vs. Ctrl	√	√
Sieh (2008)	[1]	63	TVLE (L1) vs. Ctrl	–	√
Song & Lee (2019)[e]	[1]	41	L1 vs. L2	√	√
	[2]	31	L1 vs. L2	√	√
Tian & Macaro (2012)[f]	[1]	67	L1 vs. L2	√	√
	[2]	66	TVLE (L1) vs. Ctrl	–	√
	[3]	69	TVLE (L2) vs. Ctrl	–	√
Zhang & Graham (2020)[g]	[1]	71	L1 vs. L2	√	√
	[2]	69	TVLE (L1) vs. Ctrl	√	√
	[3]	68	TVLE (L2) vs. Ctrl	√	√
Zhao & Macaro (2016)	[1]	100	L1 vs. L2	√	√
	[2]	98	TVLE (L1) vs. Ctrl	√	√
	[3]	98	TVLE (L2) vs. Ctrl	√	√
Total	36	3,304		21	34

Note. L1 = first language; L2 = second language; TVLE = teachers' verbal lexical explanation; Ctrl = control.

[a]L1 indicates that a teacher's lexical explanation was delivered in the L1, and L2 refers to teachers' explanation in the L2. Ctrl symbolizes a control condition where the same activity was conducted but without teachers' explanation

[b]Dashes indicate missing data, and ticks indicate available data.
[c]This published article was based on an unpublished doctoral dissertation (J. H. Lee, 2010).
[d]This published article was based on an unpublished master's thesis (Lugo-Neris, 2007).
[e]This published article was based on an unpublished master's thesis (Song, 2017).
[f]This published article was based on an unpublished doctoral dissertation (Tian, 2009).
[g]This published article was based on an unpublished doctoral dissertation (Zhang, 2018).

Effect Size Calculation

Calculating an effect size converts treatment effects from diverse independent samples with different scales into standardized scale-free estimates for use in meta-analysis. Table 2 shows comparisons conducted by the identified studies, such as TVLE versus control and L1 versus L2. Thus, effect sizes were calculated to measure the standardized mean differences between TVLE and control conditions, which represented the relative effects of teachers' lexical explanations (regardless of different language choices) compared to conditions. Similarly, a statistically significant positive effect size for a L1 versus L2 comparison denoted better L2 vocabulary learning through teachers' L1 input than through its L2 counterpart.

Among various effect size units, we used Cohen's unbiased *d*, also known as Hedges's *g*. To this end, we computed effect sizes by dividing the posttest mean differences by the pooled standard deviation to compare two means. Based on Cohen's *d* effect size calculation, unbiased *d* is computed by multiplying by a correction factor (*J*). The equations for the effect size calculation are shown below.

$$\text{Effect size} = J_{\text{Correction factor}} \times \frac{Mean_{\text{TVLE}} - Mean_{\text{control}}}{\sqrt{\left[\frac{(n_{\text{T}}-1)SD^2_{\text{TVLE}}+(n_{\text{C}}-1)SD^2_{\text{control}}}{n_{\text{TVLE}}+n_{\text{control}}-2}\right]}} \quad (1)$$

$$J_{\text{Correction factor}} = 1 - \frac{3}{\{4 \times (n_{\text{TVLE}} + n_{\text{control}} - 2) - 1\}} \quad (2)$$

As there were multiple independent samples in a study, as shown in Table 2, several independent samples had multiple measurements, which required us to calculate multiple effect sizes for a sample. However, to keep the multiple effect sizes for a study, we needed statistical adjustment, because multiple effect sizes for a sample may cause a dependency violation (i.e., these effect sizes came from the same sample, and thus were not independent of each other). To resolve this issue, we adopted a multilevel meta-analysis approach, which has recently gained popularity (e.g., Assink & Wibbelink, 2016; H. Lee et al., 2019, 2020), because it enables computing an overall mean effect size and conducting moderator analyses without losing the cluster membership of the calculated effect sizes (i.e., effect sizes are nested in each study). Moreover, failure to adopt such an approach would have prevented us from

analyzing the effect of the testing format, which is important to understand how the effects of TVLE were actually measured (Boers, Bryfonski, Faez, & McKay, 2021), because the testing format variable was an effect size level moderator.

Outlier Diagnostics and Publication Bias

After calculating effect sizes, we diagnosed potential outliers before further analyses. According to the suggestions made by Assink and Wibbelink (2016) and Tabachnik and Fidell (2013), we first converted our effect sizes into standardized *z* values and excluded those with absolute values greater than 3.29. We then used the metafor package (Viechtbauer, 2010) in R software to identify any outliers. As Viechtbauer and Cheung (2010) explained, this package provides studentized deleted residuals, which can be calculated by dividing the residuals of each effect size by their corresponding standard errors, along with additional measures such as DFFITS values, Cook's distances, and COVRATIO values (see Viechtbauer & Cheung, 2010, for their definitions and equations). Because this approach quantifies the results of the diagnostics, it is possible to mathematically detect outliers in a systematic manner. If an effect size is considered an outlier, it is marked with a filled circle on the figures calculated from these measures of the metafor package. For example, according to the plots in Figure 2, two effect sizes (numbers 7 and 8) were considered outliers in posttests for the TVLE versus control comparison; thus, we removed them from the dataset.

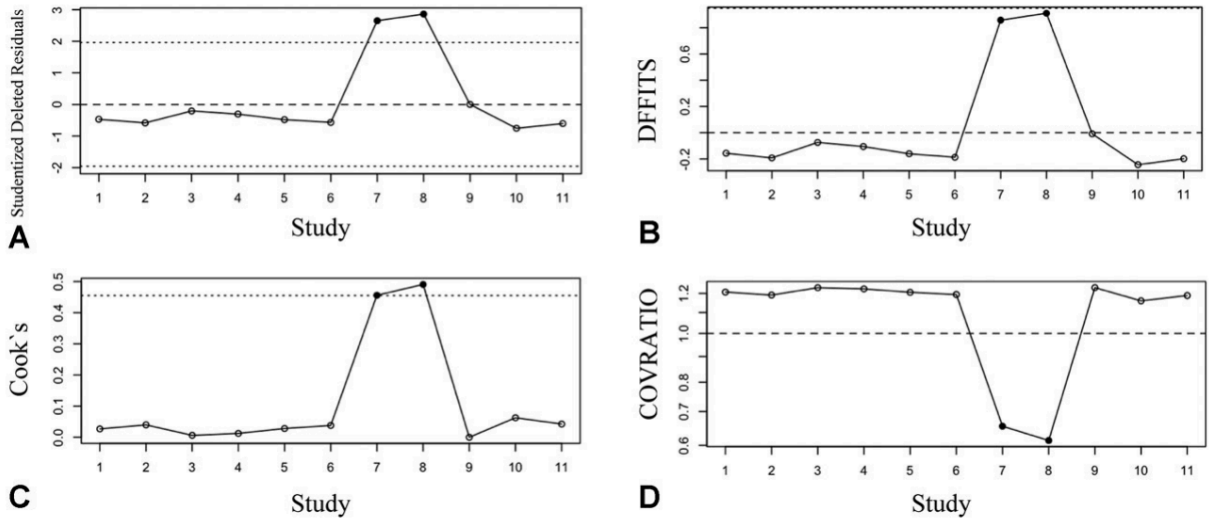


Figure 2 Plot of the (a) studentized deleted residuals, (b) DFFITS values, (c) Cook's distances, and (d) COVRATIO values for 11 effect sizes of posttests for the comparison of teachers' verbal lexical explanation versus control. Filled circles are suggested to be outliers.

Likewise, we used this approach for the remaining three datasets to exclude outliers before conducting any further data analyses. As a result, for the posttests, a total of 9 effect sizes remained from 7 samples and 15 effect sizes remained from 10 samples for the TVLE versus

control comparison and the L1 versus L2 comparison, respectively. For the delayed posttests, 22 effect sizes remained in the dataset from 15 samples and 9 effect sizes remained from 7 samples for the TVLE versus control and L1 versus L2 comparisons, respectively.

After excluding potential outliers, we checked for any signs of the small-study effect (whereby studies with small samples but with a large effect size are more frequently published than studies with large sample sizes with small effect sizes) among the calculated effect sizes. To this end, we used the metafor package to compute funnel plots and conduct Egger's bias regression test (*p*-value > .10 for no significant sign of publication bias; see the Results section).

Data Analysis Plan

We used the metafor package (Viechtbauer, 2010, Version 3.0-2) in R (Version 4.1.2) for further data analyses, including overall average effect size calculations and moderator analyses. As mentioned in the Effect Size Calculation section above, we computed multiple effect sizes according to test formats for independent samples when applicable. Because these effect sizes from the same samples were not independent of each other, their group membership should be considered when computing an overall average effect size estimate; hence, a multilevel meta-analysis (i.e., a random-effects model; Fernández-Castilla et al., 2020) was used to address this issue.

Thus, we used a three-level meta-analysis, in which the sampling variances of each effect size were at Level 1, differences between effect sizes within samples were at Level 2, and differences between samples were at Level 3 (Assink & Wibbelink, 2016). As the sampling variances of each effect size were not estimated in the meta-analysis because they had already been calculated as their standard errors, this multilevel model was also a variance-known model (H. Lee et al., 2019). When fitting a multilevel meta-analytic model with the calculated effect sizes along with their variances, an equation with only an intercept represents the overall average effect size.

In addition, moderator analyses can be performed by including them in the meta-analytic model to examine how they account for within- and/or between-sample variance. Considering our small sample size and a relevant suggestion by Hox (2010), we evaluated moderating effects separately in univariate models and then conducted a multiple moderator model only with significant moderating variables. It should be noted that any variables with a high missing-data rate were omitted for multiple regression analyses. This is because this approach is conducted by deleting individual cases from the analysis if there are missing data on any

variable in the analysis (i.e., a listwise deletion; Viechtbauer, 2010). The results of our statistical approaches were tabulated (Tables 3–5), and boldface indicates statistically significant values at $\alpha = .05$.

Finally, because there were some influential cases (i.e., cases whose exclusion from the meta-analysis may lead to considerable changes; Viechtbauer & Cheung, 2010) and a sign of publication bias (i.e., a small-study effect), we conducted a sensitivity analysis for the delayed posttests for the TVLE versus control comparison, to assess whether the inclusion of the influential cases may have affected the results of the meta-analysis.

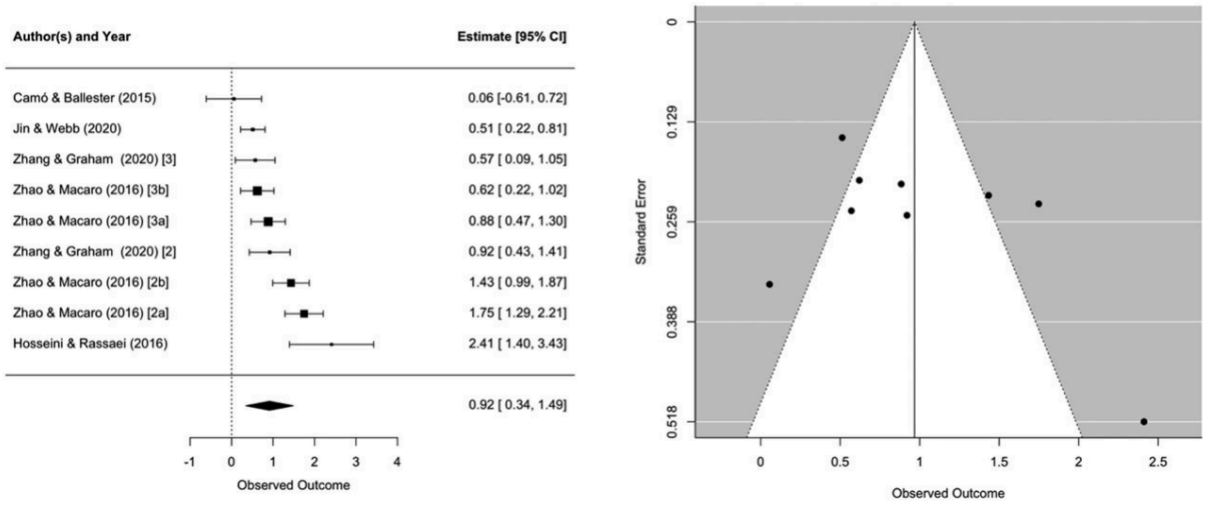


Figure 3 Forest plot (left) and funnel plot (right) for the posttest results of the comparison of teachers' verbal lexical explanation versus control.

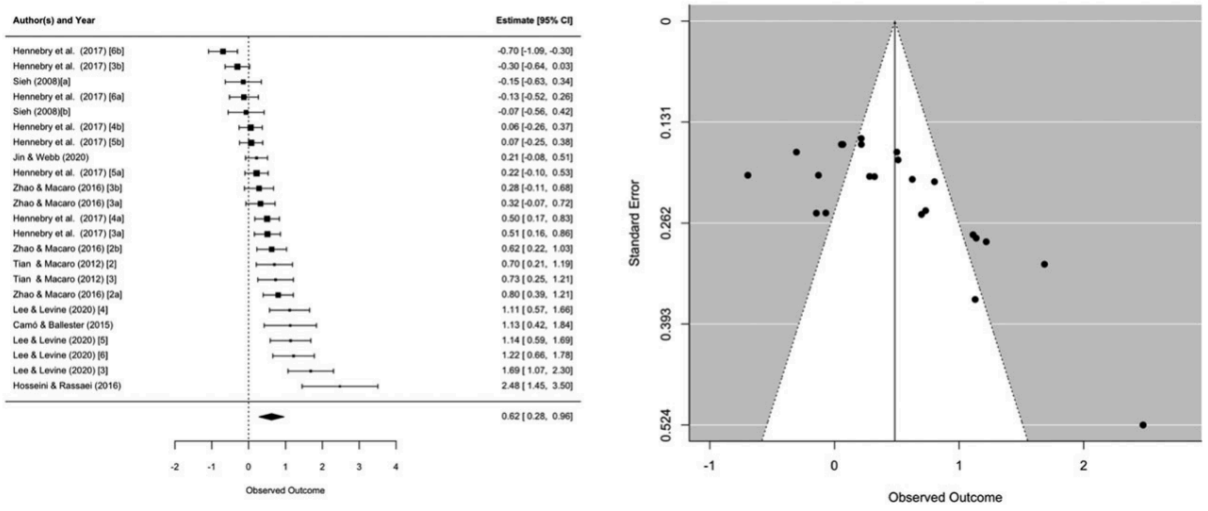


Figure 4 Forest plot (left) and funnel plot (right) for the delayed posttest results of the comparison of teachers' verbal lexical explanation versus control.

Results

This section presents the meta-analysis findings based on the posttest and delayed posttest for the TVLE versus control and L1 versus L2 comparisons to answer the research questions.

TVLE Versus Control

First, as shown in Figure 3, we found that the overall mean effect size from the posttest results of the TVLE versus control comparison was 0.92 (*SE* = 0.25, *z* = 3.69, *p* = .01, 95% CI [0.34, 1.49]) and that there was no sign of publication bias among the effect sizes (*p* value for the Egger's test = .15). Regarding the delayed posttest results (Figure 4), we found that the overall mean effect size was 0.62 (*SE* = 0.16, *z* = 3.80, *p* = .00, 95% CI [0.28, 0.96]) and that there was a clear sign of publication bias among the effect sizes (*p* value for the Egger's test = .00), highlighting the need to conduct a sensitivity analysis to confirm the robustness of the initial findings. Overall, the results based on the dataset revised by excluding influential cases identified through sensitivity analysis were similar to the initial results (see Appendix S3 in the Supporting Information online for details of the results). Taken together, the results indicated that TVLE was more effective in supporting students' L2 vocabulary development than control conditions in both the short and longer term.

Second, as shown in Table 3, there were no statistically significant differences between any moderator categories for the posttests; however, for the delayed posttests, we found that the overall effectiveness of TVLE may have been greater when it was delivered during the comprehension phase (*b* = 0.87, *SE* = 0.24, *z* = 3.64, *p* = .00, 95% CI [0.37, 1.36]), when the ratio of target words was high (*b* = 0.24, *SE* = 0.06, *z* = 3.73, *p* = .01, 95% CI [0.08, 0.39]), when the number of sessions was relatively small (*b* = −0.14, *SE* = 0.05, *z* = −3.09, *p* = .01, 95% CI [−0.24, 0.05]), or when the vocabulary test was conducted in the format of recall (*b* = 0.50, *SE* = 0.12, *z* = 4.31, *p* = .00, 95% CI [0.26, 0.75]).

Table 3 Moderator analysis for comparison of teachers' verbal lexical explanation (TVLE) versus control

Moderator variable	Categorical value	Posttest (*n* = 9; *k* = 7)						Delayed posttest (*n* = 23; *k* = 16)					
						95% CI						95% CI	
		Mean	***SE***	***z***	***p***	***LL***	***UL***	**Mean**	***SE***	***z***	***p***	***LL***	***UL***
L2 proficiency	Basic	(no baseline for comparison)						−0.78	0.64	−1.22	.24	−2.11	0.55
	Beyond basic							(reference category)					
TVLE type	Additional	−0.25	0.61	−0.41	.69	−1.69	1.19	0.10	0.52	0.20	.85	−0.98	1.19
	Definition	(reference category)						(reference category)					
Input mode	Listening	−0.87	0.39	−2.20	.06	−1.80	0.06	−0.40	0.43	−0.92	.37	−1.30	0.50
	Reading	(reference category)						(reference category)					
Timing	During	−0.36	0.57	−0.63	.55	−1.69	0.98	**0.87**	0.24	3.64	.00	0.37	1.36
	After	(reference category)						(reference category)					
Written forms	Presentation	−0.47	0.63	−0.75	.48	−1.96	1.02	−0.52	0.54	−0.96	.35	−1.65	0.61
	No presentation	(reference category)						(reference category)					
Ratio of target words	Ratio (%)	0.08	0.34	0.25	.83	−1.40	1.57	**0.24**	0.06	3.73	.01	0.08	0.39
Length of instruction (no. of sessions)	(continuous)	0.02	0.09	0.19	.85	−0.21	0.24	**−0.14**	0.05	−3.09	.01	−0.24	−0.05
Testing format	Recall	0.98	0.69	1.41	.20	−0.66	2.62	**0.50**	0.12	4.31	.00	0.26	0.75
	Recognition	(reference category)						(reference category)					
Retention interval (no. of weeks)	(continuous)	(not applicable)						0.05	0.14	0.32	.75	−0.25	0.34

Note. The level of each moderator variable set as a reference is marked as "(reference category)" in the table. Boldface indicates statistically significant values at $\alpha = .05$. L2 = second language.

When these significant moderators, except for the ratio of target words moderator, were included in a multiple moderator analysis, for the delayed posttests (see Table 4), we could identify a dominant role of the testing format, such that only this variable was significant (b = 0.46, SE = 0.12, z = 3.78, p = .00, 95% CI [0.21, 0.72]), indicating that the overall effectiveness of TVLE was greater when vocabulary knowledge was measured, at the delayed posttests, through recall tests rather than recognition tests. We omitted the ratio of words moderator, as mentioned above, due to its high missing-data rate (namely, 65%; only eight out of 23 effect sizes had values). Regarding the remaining variables, we found that the overall directions were maintained when it was delivered during the comprehension phase (b = 0.26, SE = 0.54, z = 0.49, p = .63, 95% CI [−0.87, 1.40]) and when the number of sessions was relatively small (b = −0.09, SE = 0.09, z = −1.03, p = .32, 95% CI [−0.29, 0.10]), although the results were not statistically significant.

Table 4 Multiple moderator analysis for delayed posttests from the comparison of teachers' verbal lexical explanation versus control

Moderator variable	Categorical value					95% CI	
		Mean	***SE***	***z***	***p***	***LL***	***UL***
Timing	During	0.26	0.54	0.49	.63	−0.87	1.40
	After	(reference category)					
Ratio of target words	Ratio (%)	(omitted due to missing values)					
Length of instruction (no. of sessions)	(continuous)	−0.09	0.09	−1.03	.32	−0.29	0.10
Testing format	Recall	**0.46**	0.12	3.78	.00	0.21	0.72
	Recognition	(reference category)					

Note. The level of each moderator variable set as a reference is marked as "(reference category)" in the table. Boldface indicates statistically significant values at $\alpha = .05$.

First Language Versus Second Language

From the posttest results of the L1 versus L2 comparison, we found that the overall mean effect size was 0.59 (*SE* = 0.15, *z* = 4.05, *p* = .00, 95% CI [0.27, 0.90]; Figure 5) and that there was no sign of publication bias among the effect sizes (*p* value for the Egger's test = .15). Regarding the delayed posttest results, we found that the overall mean effect size was 0.28 (*SE* = 0.13, *z* = 2.21, *p* = .04, 95% CI [0.01, 0.56]; Figure 6) and that there was no sign of publication bias among the effect sizes (*p* value for the Egger's test = .54). These results indicate that TVLE in the L1 was more effective at aiding students' L2 vocabulary learning than TVLE in the L2 in both the short and longer term. Regarding moderator analyses, as shown in Table 5, we did not find any statistically significant differences between any moderator categories.

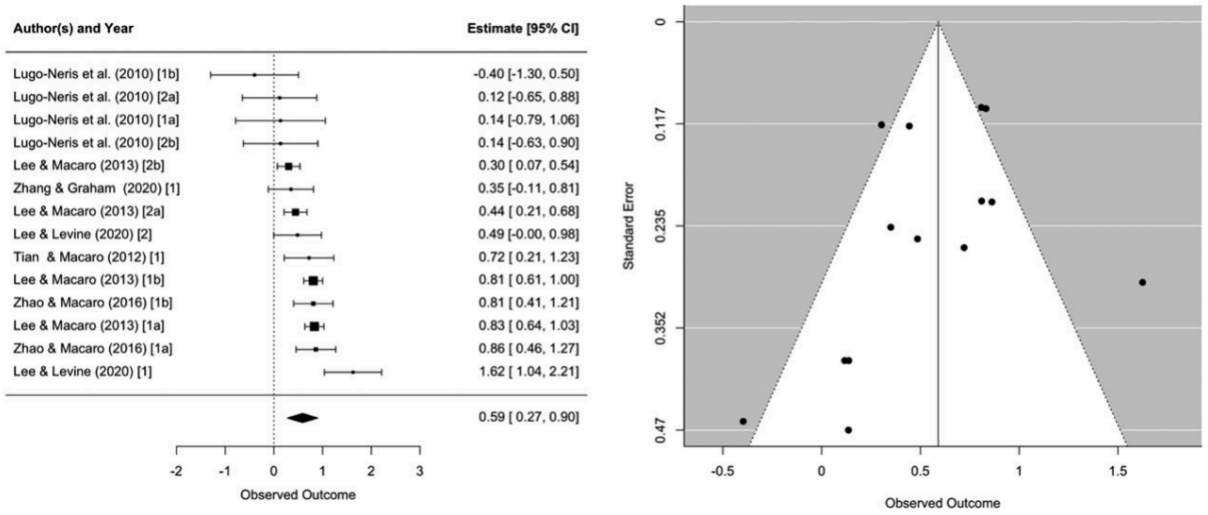


Figure 5 Forest plot (left) and funnel plot (right) for the posttest results of the comparison of first language versus second language.

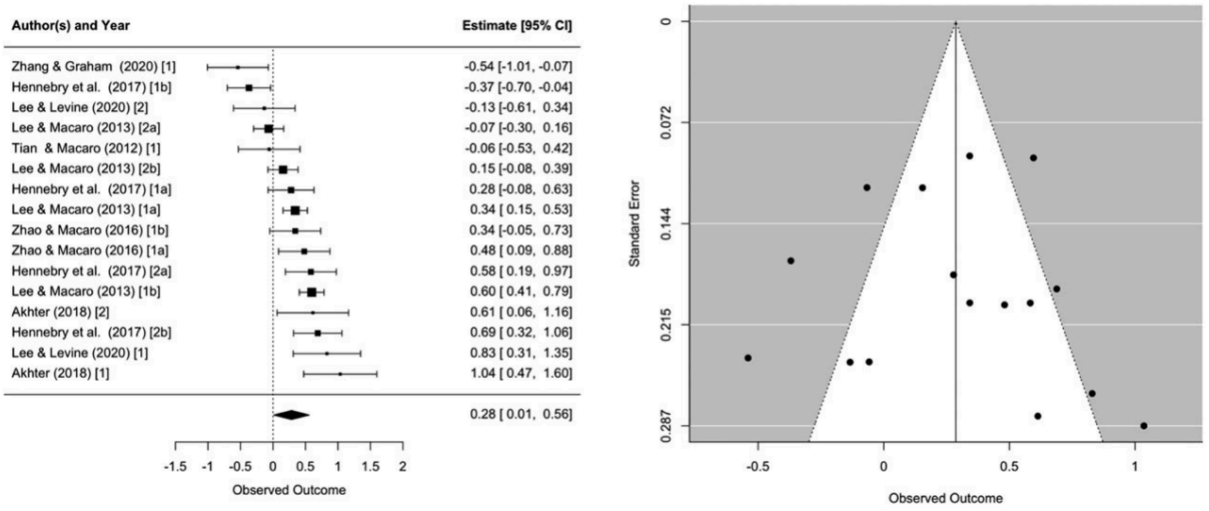


Figure 6 Forest plot (left) and funnel plot (right) for the delayed posttest results of the comparison of first language versus second language.

Table 5 Moderator analysis for comparison of first language versus second language (L2)

Moderator variable	**Categorical value**	**Posttest**						**Delayed posttest**					
						95% CI						**95% CI**	
		Mean	***SE***	***z***	***p***	***LL***	***UL***	**Mean**	***SE***	***z***	***p***	***LL***	***UL***
L2 proficiency	Basic	−0.35	0.33	−1.05	.31	−1.06	0.37	0.21	0.43	0.48	.64	−0.72	1.13
	Beyond basic	(reference category)						(reference category)					
TVLE type	Additional	−0.51	0.28	−1.83	.09	−1.12	0.10	−0.41	0.26	−1.59	.14	−0.96	0.14
	Definition	(reference category)						(reference category)					
Input mode	Listening	−0.14	0.32	−0.45	.66	−0.84	0.56	−0.33	0.26	−1.30	.22	−0.89	0.22
	Reading	(reference category)						(reference category)					
Timing	During	−0.28	0.46	−0.61	.55	−1.30	0.73	−0.36	0.25	−1.42	.18	−0.90	0.19
	After	(reference category)						(reference category)					
Written forms	Presentation	0.70	0.32	2.18	.05	0.00	1.40	(no baseline comparison)					
	No presentation	(reference category)											
Ratio of target words	Ratio (%)	0.03	0.08	0.34	.75	−0.17	0.22	0.08	0.07	1.21	.27	−0.09	0.25
Length of instruction (no. of sessions)	(continuous)	−0.08	0.04	−2.01	.07	−0.16	0.01	−0.05	0.05	−1.09	.30	−0.16	0.05
Testing format	Recall	0.11	0.10	1.02	.33	−0.12	0.33	−0.01	0.18	−0.04	.97	−0.39	0.38
	Recognition	(reference category)						(reference category)					
Retention interval (no. of weeks)	(continuous)	(not applicable)						−0.12	0.16	−0.76	.46	−0.48	0.23

Note. The level of each moderator variable set as a reference is marked as "(reference category)" in the table. TVLE = teachers' verbal lexical explanation.

Discussion

In the present meta-analysis study, we aimed to synthesize previous empirical studies on TVLE as a type of L2 lexical FonF. Our results revealed that teachers' verbal lexical FonF led to greater vocabulary gains than a lack thereof for learner groups with varying L2 proficiency levels in the context of reading or listening comprehension activities, indicating that this type of lexical FonF in meaning-focused activities would be beneficial for L2 learners' vocabulary

development. This finding concurs with those of previous studies on the effects of TVLE in L1 contexts (e.g., Brett et al., 1996; Elley, 1989) and fits with the view of L2 learning according to which the target linguistic item needs to be noticed and attended to in order for successful learning to occur (Dóczi & Kormos, 2016; Robinson et al., 2012; Schmidt, 2001). In this line of thinking, TVLE could enhance learners' attentional focus on the target words, which appears to be an important condition for these words to be "further processed in working memory" (Kormos, 2019, p. 213).

Meanwhile, such a strategical use of lexical FonF, which is momentary by nature, may enable the instructor to teach target words effectively while adhering to the general principles of meaning-focused or communicative tasks without making the target activity linguistically focused (i.e., focus on forms), although in saying this we do not mean to underestimate the value of intentional or decontextualized vocabulary learning (Webb et al., 2020). It is also possible that TVLE may further strengthen the form–meaning links of the target vocabulary, which may be beneficial for learners' subsequent L2 vocabulary learning (for relevant discussion, see the later section The Effects of Teachers' Verbal Lexical Explanation on Different Aspects of Vocabulary Knowledge).

The Effects of Language Choice in Teachers' Verbal Lexical Focus on Form

After confirming the positive effects of teachers' verbal lexical FonF, we further examined the relative effects of language choice in TVLE. The overall positive effect found in the posttest indicates that learners could make greater target vocabulary gains through teachers' L1 input than its L2 counterpart, not only in the short term, but also in the longer term. Within a limited timeframe of L2 lessons, L2 teachers often need to decide which language they should adopt for TVLE, and our findings favor the use of L1 input. These findings are in accord with predictions made by psycholinguistic models (Jiang, 2000, 2004; Kroll & Stewart, 1994) that point to the "inevitable involvement of a learner's L1 semantic system in L2 [vocabulary] learning" (Jiang, 2018, p. 24). Thus, teachers' L1 input appears to "reduce the learners' cognitive load in noticing and registering the target lexical item" (J. H. Lee & Levine, 2020, p. 266). This finding also corresponds to the result of a recent meta-analysis showing a relatively larger impact of L1 glossing than L2 glossing on L2 vocabulary learning (Kim et al., 2020).

Thus, the result regarding the benefits of L1 input over its L2 counterpart in TVLE found in the present meta-analysis may help inform pedagogical guidelines for teachers' language behaviors in meaning-focused lessons, encouraging them to briefly switch to learners' L1 from the L2—the primary medium of instruction—when opting for TVLE. From a broader

perspective, this result also lends some weight to the adoption of a bilingual approach, whose proponents have proposed that the L1 is a valuable pedagogical resource for learners and "may provide efficient shortcuts within the learning process" (Hall & Cook, 2012, p. 282).

The Effects of Teachers' Verbal Lexical Explanation on Different Aspects of Vocabulary Knowledge

Among the identified moderators, language choice (i.e., L1 or L2 input) was found to be the most important, as discussed in the previous section. Although we did not find statistically significant differences across categories for most moderators, the result regarding testing format warrants discussion, as it was the only significant finding in the multiple moderator analysis.

It has been suggested in L2 vocabulary research that there may be a hierarchy of lexical knowledge related to the form–meaning link (with recognition knowledge being acquired earlier than recall knowledge; González-Fernández & Schmitt, 2020; Laufer & Goldstein, 2004). Prior TVLE studies predominantly measured two levels of meaning knowledge, namely passive recall and passive recognition, with the former generally presupposing the latter and likely being developed at later stages (Laufer & Goldstein, 2004; Pellicer-Sánchez & Schmitt, 2010). The result of our multiple moderator analysis revealed that it was recall knowledge, rather than recognition, through which TVLE exhibited larger effect sizes than its control counterpart. One possible explanation would be that the learners assigned to the control group (i.e., lack of TVLE) may have been able to acquire some level of recognition knowledge through reading or listening to the passage only (e.g., by guessing the meaning of the target words from context), although to a lesser extent than those who received TVLE. However, recall knowledge may be more challenging to acquire without some type of instructional input (i.e., TVLE), which accounts for the superiority of the TVLE condition. In other words, TVLE may have furthered learners' L2 vocabulary knowledge, enabling learners to move from the early lexical acquisition stage of recognition knowledge to the later stage of recall knowledge.

Limitations and Future Directions

The present study is not without limitations, the most important of which may be the inclusion of only a small number of studies in the meta-analysis. Although we have followed a robust set of statistical approaches and provided qualitative analyses of the selected moderators across the studies (Appendix S1 in the Supporting Information online), we cannot exclude the possibility that the quality of our findings is limited due to the small sample size. We acknowledge that the small study sample size of the current meta-analysis is partly responsible for the relatively high standard errors (i.e., relatively low precision) of our computed average

effect size estimates for moderator analyses, and we hope that researchers will continue empirical efforts on this topic so that future meta-analysts can work with a much larger and more diverse sample of studies. Furthermore, we could not include some potential moderators due to the small sample size. For example, we did not include age as a potential moderator. However, it is possible that TVLE may have differential effects on learners of varying ages, together with other moderating variables. Overall, we believe that including larger samples in future meta-analyses would resolve these intertwined issues.

Another limitation was that the results for overall mean effect size related to the effects of teachers' lexical FonF were primarily based on "meaning" aspects of vocabulary knowledge. Although the popularity of tests on meaning aspects in these studies is expected, it is also possible that the benefits of L2 explanation for learning other aspects of L2 vocabulary knowledge, such as form or collocational knowledge, may have been obscured by this lack of diversity in testing format.

We would also encourage primary researchers to give more details about the moderator variables they include in their studies. For example, more detailed descriptions of the type of TVLE (e.g., the quantity and contents of TVLE) and the sampled students' L2 proficiency levels (see Park et al., 2021, for further discussion) would enable future meta-analysts to better capture the original variations related to these variables across the target studies. Such efforts, along with a greater number of future studies on this issue, would extend what the current meta-analysis has presented to advance the understanding of the effects of TVLE on L2 vocabulary learning.

Conclusion

The present meta-analysis sought to estimate the extent to which TVLE was effective in supporting learners' L2 vocabulary development, by systematically reviewing the body of studies that has examined its effect in the context of meaning-focused activities. The findings demonstrated that TVLE led to greater L2 vocabulary gains than a lack thereof in the short and longer term. Our analyses also detected significant effects for language choice of TVLE (i.e., with larger benefits for TVLE in the L1 than in the L2) and for the type of vocabulary knowledge (i.e., with larger effects when knowledge was measured by recall tests rather than recognition tests), but not for the other identified moderator variables (i.e., learners' L2 proficiency level, type of TVLE, input mode, timing of TVLE, availability of written forms of target words, ratio of target words to total words, and length of instruction).

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1. Qualitative Analysis of the Selected Moderators Across the Selected Studies.

Appendix S2. Summary of the Included Studies.

Appendix S3. Sensitivity Analysis for Delayed Posttests from Teachers' Verbal Lexical Explanation Versus Control Comparison.

Appendix S4. References of Included Studies.

Appendix: Accessible Summary (also publicly available at https://oasis-database.org)

Teachers' explanations about new vocabulary are effective for learning, and more effective in the first language than in the second

What This Research Was About and Why It Is Important

In instructed second language (L2) learning contexts, teachers are usually one of the most critical sources of language input. Previous studies on L2 vocabulary learning have suggested that teachers' verbal lexical explanation (TVLE) assists learners to develop L2 vocabulary knowledge more than when teachers do not provide such explanations. These studies have generally investigated L2 vocabulary learning in the context of meaning-focused activities (e.g., reading or listening comprehension), where the learning aims are to comprehend the target text, as well as to learn the words included in that text. Some of these studies have also examined whether the language of teachers' lexical explanation would make any difference, comparing L2 explanations with first language (L1) explanations. To synthesize the previous findings of these studies, we conducted a meta-analysis of 14 empirical studies selected using a careful procedure. The findings of the meta-analysis indicated that the TVLE condition was more effective than the control condition when no TVLE was provided. Our results also show that explanation in L1 was more effective than explanation in L2.

What the Researchers Did

- We screened 153 potential studies on TVLE, and, from these, selected 14 rigorous experimental studies.
- The selected studies had a combined total of 3,304 participants.
- Characteristics of the studies were coded as follows: participants' L2 proficiency; type of teachers' verbal lexical explanation; input mode (i.e., listening or reading); when the TVLE was delivered for instruction (i.e., during or after reading or listening to the target text); whether the written forms of the target words were provided; the ratio of target words to the total words of the listening or reading texts; length of instruction; and format of vocabulary tests (recognition [choose the meaning from options] or recall [produce the meaning]).
- For the subset of the studies that included the TVLE and control conditions, we calculated effect sizes by subtracting the scores in the control condition from those in the TVLE condition, such that a positive effect size indicated that providing TVLE was more effective than no TVLE.
- For the studies that compared TVLE in L1 with TVLE in L2, we calculated effect sizes by subtracting the scores in TVLE in L2 condition from those in TVLE in L1 condition, such that a positive effect size indicated that the L1 explanation was more effective than the L2 explanation.

What the Researchers Found

- Previous studies on TVLE have mainly targeted the link between the form and meaning of L2 vocabulary.
- Overall, TVLE resulted in more vocabulary gains than no explanation, both in the short- and longer-term (i.e., at posttests and delayed posttests), in the context of meaning-focused (i.e., comprehension) activities.
- TVLE in L1 was more effective than TVLE in L2 for L2 vocabulary development.
- TVLE seemed to be more effective when recall was tested than when recognition was tested.
- Other factors such as mode, when the TVLE was delivered for instruction, and length of instruction did not influence the effectiveness of TVLE.

Things to Consider

- L2 teachers should not be discouraged from explaining unknown vocabulary in their learners' L1 in the context of meaning-focused activities.

- The results of the study need to be interpreted with caution, as the total number of studies included in this meta-analysis was small (14 studies). Thus, we recommend more (quasi-) experimental studies examining the effects of TVLE and continuous meta-analytic efforts investigating TVLE.
- One potential direction for future research is to examine the effects of TVLE on other aspects of vocabulary knowledge, such as the multiple meanings of a word and its association with other words.

Materials, data, open access article: N/A

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